

1. Let n be some extremely large integer.

Let T represent a binary tree that has n nodes and satisfies the following properties:

- 1: Every node has two children unless it is a leaf.
- 2: Every node is colored either blue or orange.
- 3: If a node is blue then it can't have a blue parent.
- 4: If we follow the path from any leaf to the root (only moving upward), we must count the same number of orange nodes as in any other such path (from another leaf up to the root).

(a) Suppose that T has only orange nodes. What is its depth?

(b) What is the largest number of blue nodes that T might have?

(c) Let p be the length of the shortest path in T that begins at the root and ends at a leaf. Let q be the length of the longest root-to-leaf path in T . What is the largest possible ratio, q/p ?

(d) What is the maximum depth that a leaf might have in T ?

(e) What is the minimum depth that a leaf might have in T ?

(f) Draw a tree that satisfies the four given properties, such that the left subtree (of the root) has the smallest possible number of nodes (given the small tree size that you chose). Your example should be small, but also large enough so that we can clearly understand what your tree would look like if it were generalized to huge values of n .

(g) Suppose that you generalize your idea in (f), for huge values of n . What is the approximate ratio of the sizes of the left and right subtrees? You may use big-O.

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(b) What is the largest number of blue nodes that T might have?

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(c) What is the largest possible ratio, q/p ?

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(d) What is the maximum depth that a leaf might have in T ?

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(e) What is the minimum depth that a leaf might have in T ?

zzz

yyy

(f) Tree that minimizes left subtree size.

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(g) What is the approximate ratio of the sizes of the left and right subtrees?

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